TECHNICAL SPECIFICATION & PROTOTYPE FABRICATION BRIEF

IDLEBIN: Low-Cost Mechanical Compacting Dustbin

Category: Waste Management | Pondicherry University Grand Finale Submission

1. SYSTEM OVERVIEW & PROBLEM STATEMENT

Municipal solid waste in Indian cities and villages is dominated by high-volume, low-density plastic packaging (PET bottles, snack sachets). Uncompressed waste fills bins within hours, causing chronic street overflow and animal scattering. IdleBin provides a pure mechanical 4x volume reduction at zero electricity.

2. BILL OF MATERIALS (BOM) & PROTOTYPE COSTING

ITEM / COMPONENT	SPECIFICATION	FUNCTION	COST (INR)
Outer Body Drum	25L-30L Cylindrical HDPE Bin	Waste containment housing	**Rs. 250**
Compactor Platen	8mm Perforated Acrylic / Wood	Downwards compression plate	**Rs. 100**
Linkage Pushrod	1/2-inch Mild Steel / PVC Tube	Force transmission rod	**Rs. 120**
Foot Pedal Lever	Mild Steel Flat Bar / Re-bar	Biomechanical input pedal	**Rs. 80**
Return Springs	Dual Helical Tension Springs	Automatic plate reset	**Rs. 50**
Pivot Fasteners	M6 Galvanized Bolts + Hinges	Joint pivot points	**Rs. 40**
Bio-Filter Tray	Activated Coconut Charcoal Packet	Passive odor neutralizer	**Rs. 40**
TOTAL PROTOTYPE FABRICATION COST			**Rs. 680 (Under $9 USD)**

3. MECHANICAL ENGINEERING CALCULATIONS

Mechanical Advantage: MA = Effort Arm (Le) / Load Arm (Ll) = 27 cm / 15 cm = 1.80

Applied Force: User applies normal foot pressure F_input = ~350 N (approx 35 kg force equivalent).

Crushing Force: F_compaction = F_input * MA = 350 N * 1.80 = 630 N (Sufficient to crush 3 PET bottles simultaneously).

4. 48-HOUR RAPID PROTOTYPING PROCEDURE

1. Platen Cutting: Cut circular plate 15mm smaller than bin diameter. Drill six 12mm holes for air venting.
2. Fulcrum Mounting: Drill two 8mm pivot holes at bottom base. Mount pedal with M8 bolt and nylon lock-nut.
3. Linkage Assembly: Connect pushrod to pedal arm. Connect upper pushrod end to central platen mounting bracket.
4. Spring Tensioning: Hook dual tension springs between top rim and platen to ensure immediate recoil.
5. Testing & Validation: Conduct 50 continuous test cycles on empty PET bottles and cardboard snack boxes.

5. KEY HIGHLIGHTS FOR COMPETITION JURY

Direct SBM-U 2.0 KPI: Directly stops Garbage Vulnerable Points (GVPs) and open drain choking.

Zero Electrical Dependency: Never suffers from drained batteries, broken chips, or weather damage.

Universal Adoption: Feasible for any rural Panchayat, municipal street, school, or railway station.